Std : 8th .

Q1. Solve any four sub-questions from the following. (4mks)

1. If base of a parallelogram is 8 cm and its height is 5 cm find its area.
2. Divide $21m^2 \div 7\,m$.
3. $2m + 7 = 9$ solve the given equation.
4. If radius of a circle is 21 cm find diameter.
5. Volume of a cube is= --------.
6. Principal= 500 and Amount = 675.Find interest

1B). Solve any 2 sub-questions. (4mks)

1. $3x + 12 = 2x - 4$. Solve the equation.
2. Divide $(8p^3 - 4p^2) \div 2p^2$
3. If radius of circle is 21 cm, find area.

2A). Choose the correct alternative. (5mks)

1. The perpendicular drawn from the center of a circle to its chord _____ the chord.

   a) bisects b) equal c) congruent

2. Reduction in value of things is called_____.

   a) depreciation b) reduction c) loss d) resale

3. Average is called as _____.

   a) arithmetic mean b) geometric mean c) median

4. $40m^5 \div 30m^3$

   a) $4/3m^2$ b) $3/4m^2$ c) $4/3m^3$

5. $3x = 12$

   a) 4 b) 3 c) −4.

Q2B). Solve any 3 sub-question. (6mks)

1. 1.In a circle central angle = 90°. Find measures of major & minor arcs.
2. Area of a circular ground is 3850 sq m. Find the radius of the circular ground.

3. Lengths of the diagonals of a rhombus are 15 cm & 24 cm. Find its area.

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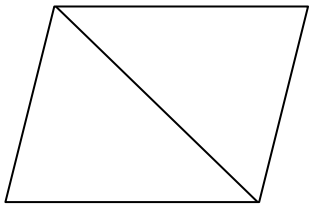
4. State the test by which two triangles in each pair diagram are congruent.



Q3A). Solve any 3 sub-questions. (6mks).

1. Find the average number of pages he read everyday from the given data.

   60, 54, 50, 46, 50.

2. Find the area of the circle if its circumference is 88 cm.
3. In a circle with center P, chord AB is drawn of length 12 cm, seg PQ ⊥ chord AB. Then find l(QB).
4. The ratio of a weight of a Copper and Zinc 13 : 7.find the weight of a zinc in a brass utensil weighing 700 gram.

Q3B). Solve any three sub-questions. (6mks)

1. Length of the two parallel sides of a trapezium are 13 cm & 9 cm respectively and its height is 8 cm. Find its area.
2. Joseph's weight is two times the weight of his younger brother. Find Joseph's weight if sum of their weight is 63 kg.
3. Solve equation: $\frac{y}{7} + \frac{y-4}{3} = 2$
4. Solve Divide : $(2y^3 + 4y^2 + 3) \div 2y^2$

Q4). Solve any 3 sub-question. (9mks)

1. The number of saplings planted by students:

   Fill in the boxes and find average number of saplings planted by each student.

Number of Sapling(xi)	**Number of student**	**fixi**
1	4	4
2	6	□

3	12	□
4	8	32
	N = □	∑fi xi = □

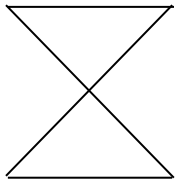
Mean ($\bar{x}$) = $\frac{\square}{N}$

$= \frac{84}{\square}$

$= \square$

2. In the adjacent figure seg AD ~ seg EC which additional information is needed to

   Show that ΔABD and ΔABC will be congruent by A. A. S test.



3. A loan of Rs 15,000 was taken on compound interest. If the rate of compound interest is 12% p.c.p.a. Find the amount to settle the loan after 3 years.

4. Find the total surface area of a closed cylindrical drum if its diameter is 50 cm and height is 45 cm. (π = 3.14).

Q5) .Solve any one **(4mks)**

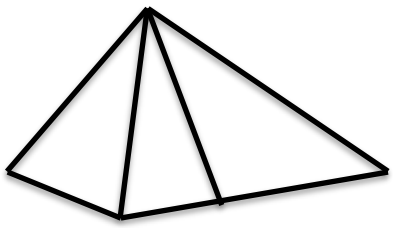
1. The Cost Price of a machine is 2,50,000. If the rate of depreciation is 10% per year, find its depreciation in value of the machine after 2 years.
2. Divide $(4x^4 - 5x^3 - 7x + 1) \div (4x - 1)$

Q6). Solve any 2. **(6mks)**

1. Draw the sub-divided bar graph.

Town	Pune	Mumbai	Kolhapur
Female	120	40	60
Male	180	80	100

2. Some measures are given in the adjacent figure. Find the area of □ ABCD.



3. A zinc sheet is of length 3.3 m & breadth 3 m. How many pipes of length 30 cm and radius 3.5 cm can be made from this sheet?